Motivation Letter

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My research interests lie at the intersection of robot learning, perception, and manipulation, particularly in developing systems that can translate high-level task descriptions into robust robot behavior. In recent work at KTH, Ericsson Research, and currently ABB Corporate Research, I have been exploring problems related to language grounding, structured representations, and learning based methods (specifically Reinforcement Learning and Diffusion Models) for robotic control. These experiences have shaped my interest in scalable robot learning systems that combine multimodal perception with structured reasoning and control. The focus of this PhD project on VLA models and on leveraging whole-body motion to improve perception and manipulation closely aligns with the research direction I aim to pursue during doctoral studies.

Prior to joining KTH, I completed a dual degree at the Indian Institute of Technology Madras (B.Tech in Engineering Design and M.Tech in Robotics), where I developed strong foundations in robotics, control, and machine learning. During this time I worked on several robotics systems projects spanning perception, planning, and control. In one project I developed a perception pipeline using structure-from-motion to reconstruct human torso geometry from monocular RGB images for robotic ultrasound positioning. This system integrated COLMAP-based reconstruction with ROS and MoveIt to generate manipulation trajectories for a simulated ultrasound probe. These early projects introduced me to the full robotics stack and motivated my interest in building intelligent systems that can autonomously interpret and interact with their environments. I was also awarded the IIT Madras Young Research Fellowship in recognition of strong research potential.

Since joining KTH, I have focused on learning-based approaches for robotic perception and manipulation. I joined the Robotics, Perception and Learning (RPL) group through the research project course, where I worked on a year-long project on deformable object manipulation under the supervision of Prof. Florian Pokorny. In this work I explored how vision-language models can help interpret manipulation scenes and translate high-level instructions into structured robot-executable representations. I developed a taxonomy-guided prompting framework that mapped natural language descriptions to motion primitives and evaluated failure modes in the system to improve robustness and interpretability. This work resulted in a workshop submission to ROMADO at IROS 2025. More importantly, the project exposed me to the full research cycle: identifying open questions, designing experiments, implementing systems, and communicating results.

In parallel, I worked as a research intern at Ericsson Research in Lund, where I focused on developing robust 3D scene graph representations for real-world environments. My work integrated instance segmentation, tracking, and RGB-D inpainting into a state-of-the-art scene graph pipeline to enable privacy-aware scene reconstruction and improved robustness to parameter sensitivity. I analyzed several scene graph frameworks to identify research gaps in object merging, parameter robustness, and real-time deployment. This work resulted in a first-author paper accepted at IMPROVE 2026 and another submission currently under review at ICPR 2026. Through this experience I gained valuable insight into collaborative research within an industrial setting and the peer-review process for international conferences.

I am currently conducting my Master's thesis at ABB Corporate Research in Västerås, where I investigate hybrid approaches that combine formal specification languages with learning-based control for robotic manipulation. In particular, my work explores possible approaches of integrating Signal Temporal Logic (STL) with learning based methods (Reinforcement Learning and Flow matching models) to translate high-level task specifications into safe control policies for industrial manipulation tasks.

Alongside my research work, I had taken advanced coursework like Robot Learning and Embodied AI and Deep Learning Advanced to deepen my understanding of modern learning-based robotics methods. As part of these courses I implemented semantic object mapping pipelines using open-vocabulary vision models and experimented with Vision-Language-Action models for robotic manipulation. Through these experiments I studied the generalization limits of current VLA architectures on unseen tasks, which further motivated my interest in developing architectures that combine large multimodal models with structured control frameworks.

Across these projects, a common theme in my research has been enabling robots to translate high-level instructions into reliable manipulation behavior. I am particularly interested in how large multimodal models can provide semantic grounding while structured representations and formal control frameworks ensure safety, interpretability, and robustness. I believe progress in robot manipulation will require this combination of scalable learning with structured reasoning, especially as robots move toward operating in open-ended environments.

Through my work at RPL, Ericsson Research, and ABB Corporate Research, I have developed a research approach that combines independent exploration with structured collaboration. I enjoy tackling open-ended problems, iterating through experiments, and translating theoretical ideas into practical implementations on robotic systems. Having already experienced the collaborative and intellectually stimulating environment at RPL, I believe the robotics research community at KTH would provide an excellent setting to pursue doctoral research and contribute to advances in robot learning for manipulation. Thank you for considering my application.